# The empirical recurrence for OEIS A250978: recovering a conjecture that is not written down

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**Abstract**

OEIS A250978 records “Empirical recurrence of order 97”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 243$ vertices and satisfies some monic recurrence of order at most $S$; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 97, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 97 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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## 1 A conjecture one cannot read

OEIS A250978 is “Number of (n+1)X(4+1) 0..2 arrays with no 2X2 subblock having the sum of its diagonal elements less than the absolute difference of its antidiagonal elements.”. It has offset 1 and begins

$$35381, 5598430, 883387252, 139471732486, 22020178055066, 3476591604928509, 548891857207523016, 866602721$$

The entry states:

> Empirical recurrence of order 97 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 12:56:19, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

## 2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining $u$ to $v$ when $v$ may follow $u$, and an admissible array is exactly a walk. Hence, with $M$ the adjacency matrix on $S = 243$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing $j$ fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem $M^S$ is an integer combination of $I, M, \ldots, M^{S-1}$, so:

**Lemma 1.** *$a$ satisfies a monic linear recurrence with integer coefficients of order at most $S = 243$.*

# 3 Recovering the recurrence

**Lemma 2.** *A sequence known to satisfy some linear recurrence of order at most $S$ is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 486$ exact terms of the model returns order 97 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{97} c_i\, a(n-i),$$

| | | | | | |
|---|---|---|---|---|---|
| $c_1$ | 243 | $c_{26}$ | 9402023712760367892280033 34 | $c_{51}$ | 51149775147697835995940 |
| $c_2$ | −17987 | $c_{27}$ | −5540079278777886230712815868 | $c_{52}$ | −5060168092651509077538 |
| $c_3$ | 904219 | $c_{28}$ | 2126923568449687128517465941 0 | $c_{53}$ | 46258824939429224734951 |
| $c_4$ | −35217078 | $c_{29}$ | −6348889248760803726681589537 4 | $c_{54}$ | −4374818067271009302438 |
| $c_5$ | 1065546283 | $c_{30}$ | 1541715717037184287387799447 50 | $c_{55}$ | 44697804165293513691575 |
| $c_6$ | −25672306370 | $c_{31}$ | −3049344448837320208717527592 04 | $c_{56}$ | −4573235396535518692775 |
| $c_7$ | 499921278745 | $c_{32}$ | 4720928633887698271813145847 43 | $c_{57}$ | 42675038595314084103539 |
| $c_8$ | −7840844812250 | $c_{33}$ | −4930786116514917278758472914 87 | $c_{58}$ | −3504614955510789341750 |
| $c_9$ | 97146146652644 | $c_{34}$ | 6753902133844953884053980391 2 | $c_{59}$ | 25470054765308262472998 |
| $c_{10}$ | −900041717343031 | $c_{35}$ | 1106065404726689114809430491 628 | $c_{60}$ | −1633973550853854872700 |
| $c_{11}$ | 5110600719870636 | $c_{36}$ | −2985706345223500874646159385 690 | $c_{61}$ | 8834038015140756442384 |
| $c_{12}$ | 6790385954554485 | $c_{37}$ | 4807953652954531066776810813 797 | $c_{62}$ | −3651355997971389558832 |
| $c_{13}$ | −587209602994957590 | $c_{38}$ | −5118627880435789973398380143 884 | $c_{63}$ | 9305761404419027459469 |
| $c_{14}$ | 8419305608614319947 | $c_{39}$ | 2599684300066972216538790980 822 | $c_{64}$ | 216281505718876402299 |
| $c_{15}$ | −7943567277717942696 2 | $c_{40}$ | 2610879952904103318156854362 716 | $c_{65}$ | −1693946567937726613 53 |
| $c_{16}$ | 55889232803080453629 2 | $c_{41}$ | −8305927647523301802311301024 616 | $c_{66}$ | 1108767558364621633845 |
| $c_{17}$ | −29327773354546127001 15 | $c_{42}$ | 1132362787854810455163721295 1202 | $c_{67}$ | −442453779153377953350 |
| $c_{18}$ | 10122401002866169124959 | $c_{43}$ | −9904523197476870852493982788 743 | $c_{68}$ | 3654918851223685666555 |
| $c_{19}$ | −55407902430352269938 39 | $c_{44}$ | 5335110572510276334402790736 720 | $c_{69}$ | 113605605199815646533 |
| $c_{20}$ | −21950741692255221042 6874 | $c_{45}$ | −7796677661138206870884277068 33 | $c_{70}$ | −106426620482854208 26 |
| $c_{21}$ | 18936990015961677599812 89 | $c_{46}$ | −2327803388611231803403461959 133 | $c_{71}$ | 5584619243932581901 28 |
| $c_{22}$ | −97815373036822374927540 05 | $c_{47}$ | 6454241264429257208351036337 436 | $c_{72}$ | −208672256642291666669 |
| $c_{23}$ | 35191494032982757789902 049 | $c_{48}$ | −1545875476027226828178690382 3531 | $c_{73}$ | 62443712585477678177 |
| $c_{24}$ | −79772477658228693882231 350 | $c_{49}$ | 2955681976716880938620740280 0406 | $c_{74}$ | −1437571928135219742 |
| $c_{25}$ | 98886961068894557421533 14 | $c_{50}$ | −4358676351814763486926216192 1731 | $c_{75}$ | 3893604600606737859 |

**Theorem 3.** *The recurrence above holds for every $n$, and it is the recurrence the entry records.*

*Proof.* It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 97, possibly only from some index on. Let $q$ be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $97 + 4$ terms removed returns order 97 as well, so $\deg q_{\min} = 97 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

# 4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $97 + 4$ terms removed, which is what the identification above requires. All three agree.

**Remark 4.** What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 97 that this sequence satisfies — and that family turns out to have exactly one member.

## References


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